1 Why distribution-valued time series matter

Many real-world systems evolve as probability distributions on the simplex, not single scalars.

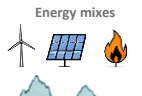
Queue occupancies



Mobility shares



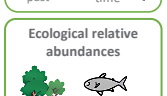
Energy mixes



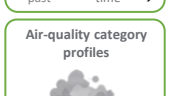
Public-health mixtures



Ecological relative abundances



Air-quality category profiles

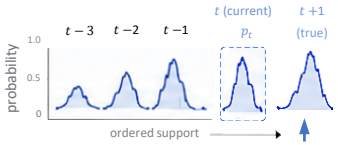


The target is an evolving probability distribution on an ordered or categorical support.

2 The challenge: aliasing

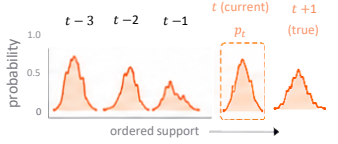
The same current distribution can imply different futures under different histories.

History A: entering congestion



Same current state p_t (same distribution)

History B: draining after a burst

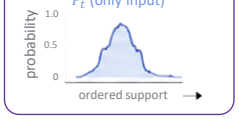


Latent transition-kernel aliasing: same p_t, different true next distributions.

3 Why standard forecasters fail

Current-only or fixed-summary models average incompatible successors.

Current-only or fixed-summary model



Averaged successor (predicts the mean)



Valid distribution, wrong forecast.

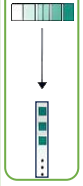
It misses regime and produces a blurred, unbiased guess.

4 CAST: structured prediction pipeline

Leverages context, similar histories, persistence, and local transport.

1 Causal context encoder

history window (past d steps)



selects the right regime

2 Retrieve Empirical successors r_t

memory retrieval



top-k similar successors

stabilizes with persistence

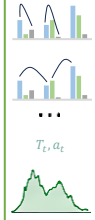
3 Persistence anchor a_t



$a_t = \lambda_t p_t + (1-\lambda_t) r_t$

4 Bounded local transport T_t

move mass neighborhood



moves mass locally on ordered support

5 Forecast $P_t + 1$

predicted next-step distribution



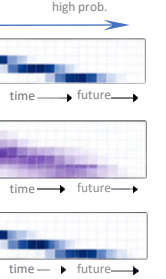
produce forecast

$$\hat{p}_{t+1} = (1-\rho_t)a_t + \rho_t T_t a_t$$

Simplex-valid by construction

5 Why this leads to better performance

Especially strong on ordered supports (e.g., queue occupancies over time)



resolves regime aliasing

preserves local mass motion

stable one-step and rollout forecasts

Best average rank across 11 benchmark sections; top-1 on 8/11 for offline KL and 8/11 for rollout JSD.